Integration: Indefinite Integrals

Questions 1–40

Question 1. Find

a) $\int (8x + 1)\, dx$

b) $\int (6x^2 - 3)\, dx$

c) $\int (10x^3 + 9)\, dx$

d) $\int (3x + 5)(4x^2 - 1)\, dx$

Question 2. Find

a) $\int \left(\frac{2}{3}x + \frac{5}{3}x^4\right)\, dx$

b) $\int \left(\frac{3}{4}x^2 - \frac{14}{3}x^6\right)\, dx$

c) $\int \left(\frac{2}{5}x^7 - \frac{3}{8}x^5\right)\, dx$

d) $\int \left(\frac{1}{3}x^2 + 4\right)\left(\frac{2}{3}x^2 + 6\right)\, dx$

Question 3. The curve y has gradient function $\frac{dy}{dx} = 2-2x$. Sketch the family of solutions that solve this differential equation.

Question 4. The curve y has gradient function $\frac{dy}{dx} = \left(\frac{x}{2} - 1\right)\left(\frac{x}{3} + 1\right)$ and passes through the point $(6, 1)$. Show that y can be written in the form $y = ax^3 + bx^2 + cx + d$, where a, b, c and d are rational numbers.

Question 5. Find the equation of the curve y given the gradient function is:

a) $\dfrac{dy}{dx} = x^{2/3}$

b) $\dfrac{dy}{dx} = 6x^{1/5}$

c) $\dfrac{dy}{dx} = 21x^{3/4}$

d) $\dfrac{dy}{dx} = (2x^{1/6} + 3)^2$

Question 6. Find

a) $\displaystyle\int (x^3 + x^{3/2})\,dx$

b) $\displaystyle\int (5x^{-2} - 2x^{2/3})\,dx$

c) $\displaystyle\int (8x^{-3} - 4x^{6/5})\,dx$

d) $\displaystyle\int (2x^{-4} - 3x^{4/3})^2\,dx$

Question 7. Find $\int (1 - x^{-2})(1 - x^{-4})\, dx$

Question 8. The curve y has gradient function $\dfrac{dy}{dx} = 40x^{1/3} - 4x$ and passes through the point $(1, -1)$. Show that y can be written in the form $y = ax^{4/3} + bx^2 + c$, where a, b and c are integers.

Question 9. Find the equation of the curve y given the gradient function is:

a) $\dfrac{dy}{dx} = 5 - 2x^{-4}$

b) $\dfrac{dy}{dx} = 6x^{-3}(x - 2)$

c) $\dfrac{dy}{dx} = x^{1/2}(8x - 3)$

d) $\dfrac{dy}{dx} = (2x^{3/2} - 3)(2x^{-7/2} + 7)$

Question 10. Find

a) $\displaystyle\int 4\sqrt{x}(x - 1)\,dx$

b) $\displaystyle\int \sqrt[4]{x}\,dx$

c) $\displaystyle\int \sqrt[5]{32x}\,dx$

d) $\displaystyle\int \sqrt[4]{81x^{1/4}}\,dx$

Question 11. The curve y has gradient function $\frac{dy}{dx} = x^2(7\sqrt{x} - 3)$ and passes through the point $(1, -4)$. Show that y can be written in the form $y = ax^{7/2} - bx^3 + c$, where a, b and c are integers.

Question 12. Find the equation of the curve y given the gradient function is:

a) $\frac{dy}{dx} = \frac{4}{x^2}$

b) $\frac{dy}{dx} = \frac{6}{x^3}$

c) $\frac{dy}{dx} = \frac{9}{5x^4}$

d) $\frac{dy}{dx} = \frac{100}{9x^{10}}$

Question 13. Find

a) $\int \frac{4}{3\sqrt{x}}\,dx$

b) $\int \frac{6}{\sqrt[3]{x}}\,dx$

c) $\int \left(\frac{8}{\sqrt[5]{x}} - \frac{100}{\sqrt[3]{8x}}\right)\,dx$

d) $\int \left(\frac{1}{\sqrt[3]{x}} - \frac{3}{x\sqrt{x}}\right)^2\,dx$

Question 14. The curve y has gradient function $\frac{dy}{dx} = \frac{8}{\sqrt{x}}$ and passes through the point $(16, -8)$. Show that y can be written in the form $y = a(2\sqrt{x} + b)$, where a and b are integers.

Question 15. The curve y has gradient function $\frac{dy}{dx} = \frac{750}{x^4}$ and passes through the point $(5, 248)$. Show that y can be written in the form $y = a\left(1 - \frac{1}{x^3}\right)$, where a is an integer.

Question 16. The curve y has gradient function $\frac{dy}{dx} = \frac{6}{x\sqrt{x}}$ and passes through the point $(8, 0)$. Show that y can be written in the form $y = a\left(\sqrt{b} + \frac{c}{\sqrt{x}}\right)$, where a, b and c are integers.

Question 17. The curve y has gradient function $\frac{dy}{dx} = \frac{12}{\sqrt[3]{x}}$ and passes through the point $(27, 0)$. Show that y can be written in the form $y = a(x^{2/3} + b)$, where a and b are integers.

Question 18. The curve y has gradient function $\frac{dy}{dx} = \frac{1}{x^3\sqrt{x}}$ and passes through the point $\left(1, \frac{3}{5}\right)$. Show that y can be written in the form $y = a - \frac{b}{cx^2\sqrt{x}}$, where a, b and c are integers.

Question 19. Find $\int \left(\frac{4}{x^2\sqrt{x}} - \frac{\sqrt{x}}{4x^2} \right) dx$

Question 20. Find $\int \left(\frac{6}{5x^3\sqrt{x}} - \frac{\sqrt[3]{x}}{6x^3} \right) dx$

Question 21. Find $\int \left(\frac{3\sqrt{x}}{4x^4} - \frac{x\sqrt{x}}{\sqrt[3]{x}} \right) dx$

Question 22. It is given that $f'(x) = \left(3x - \frac{4}{x}\right)^2$ and $f(2) = -2$. Show that $f(x) = ax^3 + bx + \frac{c}{x} + d$ where a, b, c and d are integers.

Question 23. It is given that $k > 0$. Find the equation of the curve y given the gradient function is:

a) $\dfrac{dy}{dx} = kx^4$

b) $\dfrac{dy}{dx} = (kx)^4$

c) $\dfrac{dy}{dx} = \dfrac{1}{kx^4}$

d) $\dfrac{dy}{dx} = \dfrac{18}{(2kx)^5}$

Question 24. Find

a) $\displaystyle\int \frac{5 + x^3}{x^2}\, dx$

b) $\displaystyle\int \frac{3x - 2x^3}{x^5}\, dx$

c) $\displaystyle\int \frac{8x^2 - 9}{x^4}\, dx$

d) $\displaystyle\int \frac{30x^3 - 7}{14x^5}\, dx$

Question 25. Find the equation of the curve y given the gradient function is:

a) $\dfrac{dy}{dx} = \dfrac{5x^4 - 7}{5x^3}$

b) $\dfrac{dy}{dx} = \dfrac{3 - 8x^3}{4x^5}$

c) $\dfrac{dy}{dx} = \dfrac{6x^7 - 8}{12x^4}$

d) $\dfrac{dy}{dx} = \dfrac{8x - 3x^2 - 6}{24x^5}$

Question 26. It is given that $f'(x) = \dfrac{x - 8}{x^3}$ and $f(1) = 0$. Show that $f(x) = \dfrac{ax^2 + bx + c}{x^2}$ where a, b and c are integers.

Question 27. It is given that $f'(x) = \left(\dfrac{14-7x}{\sqrt[3]{x}}\right)^2$ and $f(1) = 462$. Show that $f(x) = 21x^{1/3}(ax^2+bx+c)$ where a, b and c are integers.

Question 28. Prerequisite knowledge for Q28: Stationary Points

A curve has gradient function $\dfrac{dy}{dx} = 3x^2 + kx - 21$. The curve has two turning points. One of the turning points is at $(1, -6)$. Find the coordinates of the other turning point.

Question 29. Prerequisite knowledge for Q29–Q30: The Factor Theorem
A curve has gradient function $f'(x) = kx^2 + 74x + 11$ where $f(-1) = -100$. It is also given that $(x + 5)$ is a factor of $f(x)$. Find $f(x)$ in its simplest form.

Question 30. A curve has gradient function $f'(x) = 576x^3 + ax^2 - 208x + 40$. It is given that $(6x - 5)$ and $(2x + 1)$ are factors of $f(x)$. Find $f(x)$ in its simplest form.

Question 31. Prerequisite knowledge for Q31: Binomial Expansion
Find

a) $\int [(5+x)^4 - (5-x)^4]\,dx$

b) $\int [(3+2x)^5 - (3-2x)^5]\,dx$

c) $\int [(1-3x)^6 - (1+3x)^6]\,dx$

d) $\int [(1+2x)^7 - (1-2x)^7]\,dx$

Question 32. The curve y has gradient function $\dfrac{dy}{dx} = (4+2x)^3 - (4-2x)^3$ and passes through the point $(-2, 1024)$. Show that y can be written in the form $y = 4(x^2 + a)^2$, where a is an integer.

Question 33. Prerequisite knowledge for Q33: Second Derivatives
Find the equation of the curve y given:

a) $\frac{d^2y}{dx^2} = 3$

b) $\frac{d^2y}{dx^2} = 4x - 5$

c) $\frac{d^2y}{dx^2} = 6x^2 - 8x + 9$

d) $\frac{d^2y}{dx^2} = 8x^3 - 7x^5$

Question 34. Find the equation of the curve y given:

a) $\frac{d^2y}{dx^2} = \sqrt{x}$

b) $\frac{d^2y}{dx^2} = \frac{5}{\sqrt{x}}$

c) $\frac{d^2y}{dx^2} = \frac{3}{x^3} - \frac{x^3}{3}$

d) $\frac{d^2y}{dx^2} = \frac{8}{x^4\sqrt{x}}$

Question 35. Find the equation of the curve y given $\dfrac{d^3y}{dx^3} = \dfrac{10}{x\sqrt{x}}$.

Question 36. It is given that the curve $y = f(x)$, where $f(-3) = 2$ and $f'(3) = 2$. It is also given that $f''(x) = 14 - 6x$. Find the equation of the curve $y = f(x)$ in its simplest form.

Question 37. It is given that the curve $y = f(x)$, where $f(1) = 3$ and $f'(1) = 5$. It is also given that $f''(x) = 36x^2 + 24x + 12$. Find the equation of the curve $y = f(x)$ in its simplest form.

Question 38. It is given that the curve $y = f(x)$, where $f(2) = 5$ and $f'(1) = -2$. It is also given that $f''(x) = \dfrac{18}{x^3}$. Find the equation of the curve $y = f(x)$ in its simplest form.

Question 39. It is given that the curve $y = f(x)$, where $f(1) = 11$ and $f'(0) = 0$. It is also given that $f''(x) = 210(x - \sqrt{x} + x\sqrt{x})$. Find the equation of the curve $y = f(x)$ in its simplest form.

Question 40. Prerequisite knowledge for Q40: Stationary Points
It is given that the curve $y = f(x)$, where $f(2) = -37$ and $f'(1) = -15$. It is also given that $f''(x) = 12(kx - 1)$ and the curve has a stationary point at $x = \frac{1}{2}$. Find the equation of the curve $y = f(x)$ in its simplest form.